Intervals of Convergence, Endpoint by Endpoint
Answer Key
AP Calculus BC

Quick Answers

1. the radius = ∞, the sum at x = 2 = e^2, —
2. the radius = 3, the sum at x = 1 = $\ln\left(\frac{3}{2}\right)$, —
3. the radius = 1, the sum at the right endpoint = $\frac{\pi^2}{6}$, —
4. the radius = 0, —
5. the radius = 2, the sum at the right endpoint = $-\ln(2)$, —
6. the radius = 1, the limit of the endpoint terms = 0, —
7. the radius = $\frac{1}{3}$, the sum at x = 11/5 = $\ln\left(\frac{5}{2}\right)$, —
8. the radius = 5, the value at a dose of 6 mg = $\ln\left(\frac{5}{3}\right)$, —

What is verified

15 of 23 answers machine-checked against SymPy, across 8 problems.
— marks an answer only you can judge — problems 1, 2, 3, 4, 5, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus BC
LIM-8 – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–5 of 5 across 23 tagged checks.

1. $\sum_{n\geq 0} x^n/n!$, coefficient ratio $n+1$. (a) Radius. (b) Sum at $x=2$. (c) Interval.

Solution (a): $R = \lim_{n\to\infty} |c_n/c_{n+1}| = \lim(n+1)$, which grows without bound. $\boxed{R=\infty}$

Solution (b): This is the Maclaurin series for e^x, so at $x=2$ the sum is $\boxed{e^2 \approx 7.389}$

Solution (c) — instructor-judged. With $R=\infty$ the interval of convergence is $(-\infty,\infty)$: there are no endpoints, because no finite value of x ever sits on the boundary. The ratio computation shows why — the ratio $|x|/(n+1)$ tends to 0 for every fixed x. Full credit ties the answer to that computation.

2. $\sum_{n\geq 1} x^n/(n3^n)$, coefficient ratio $3(n+1)/n$. (a) Radius. (b) Sum at $x=1$. (c) Interval.

Solution (a): $R = \lim \dfrac{3(n+1)}{n}$. $\boxed{R=3}$

Solution (b): $\sum_{n\geq 1} t^n/n = -\ln(1-t)$; at $x=1$ the terms are $(1/3)^n/n$, so the sum is $-\ln\left(1-\frac{1}{3}\right)$. $\boxed{\ln\frac{3}{2} \approx 0.405}$

Solution (c) — instructor-judged. Centre 0, radius 3, so test $x=3$ and $x=-3$. At $x=3$ the series is $\sum\frac{1}{n}$, the harmonic series, which diverges. At $x=-3$ it is $\sum\frac{(-1)^n}{n}$, the alternating harmonic series, which converges. The interval is $[-3,3)$. Full credit names both series.

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3. $\sum_{n\geq 1}(x-2)^n/n^2$, coefficient ratio $(n+1)^2/n^2$. (a) Radius. (b) Sum at the right endpoint. (c) Interval.

Solution (a): $R = \lim\dfrac{(n+1)^2}{n^2}$. $\boxed{R=1}$

Solution (b): The centre is 2, so the right endpoint is $x=3$ and the series becomes $\sum\frac{1}{n^2}$, whose sum is $\boxed{\frac{\pi^2}{6}}$

Solution (c) — instructor-judged. At $x=3$ the series is the convergent p-series with $p=2$. At $x=1$ it is $\sum\frac{(-1)^n}{n^2}$, which converges (absolutely, by the same p-series). Both endpoints converge, so the interval is $[1,3]$ — square brackets at both ends. Full credit names the test used at each end.

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4. $\sum_{n\geq 0} n!\,x^n$, coefficient ratio $1/(n+1)$. (a) Radius. (b) Interval and behaviour elsewhere.

Solution (a): $R = \lim\dfrac{1}{n+1}$. $\boxed{R=0}$

Solution (b) — instructor-judged. A radius of 0 means the series converges only at its centre, $x=0$, where every term after the first is zero. The interval of convergence is the single point $\{0\}$; there are no endpoints to test. For any $x\neq 0$ the ratio $|x|(n+1)$ grows without bound, so the terms do too and the series diverges. Full credit gives both halves.

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5. $\sum_{n\geq 1}(-1)^n(x+1)^n/(n2^n)$, coefficient ratio $2(n+1)/n$. (a) Radius. (b) Sum at $x=1$. (c) Interval.

Solution (a): $R = \lim\dfrac{2(n+1)}{n}$. $\boxed{R=2}$

Solution (b): At $x=1$ we have $x+1=2$, so the terms become $\dfrac{(-1)^n 2^n}{n\,2^n} = \dfrac{(-1)^n}{n}$, the alternating harmonic series. $\boxed{-\ln 2 \approx -0.693}$

Solution (c) — instructor-judged. The powers are of $(x+1)$, so the centre is $x=-1$ and the endpoints are $x=1$ and $x=-3$. At $x=1$ the series is the alternating harmonic series and converges. At $x=-3$ we have $x+1=-2$, and the two sign factors cancel: the series becomes $\sum\frac{1}{n}$ and diverges. The interval is $(-3,1]$. Full credit shows that cancellation.

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6. $\sum_{n\geq 1} x^n/\sqrt{n}$, coefficient ratio $\sqrt{n+1}/\sqrt{n}$. (a) Radius. (b) Limit of $1/\sqrt{n}$. (c) Why the endpoints differ.

Solution (a): $R = \lim \dfrac{\sqrt{n+1}}{\sqrt{n}}$. $\boxed{R = 1}$

Solution (b): The term sizes at either endpoint are $\frac{1}{\sqrt{n}}$, whose limit is $\boxed{0}$

Solution (c) — instructor-judged. At $x = 1$ the series is $\sum \frac{1}{\sqrt{n}}$, the p-series with $p = \frac{1}{2} \leq 1$, which diverges even though its terms shrink to zero. At $x = -1$ it is $\sum \frac{(-1)^n}{\sqrt{n}}$, whose term sizes decrease to zero, so the alternating series test gives convergence. Same sizes, different verdicts, because alternation lets cancellation do work that a positive series cannot. The interval is $[-1, 1)$. Full credit gives both tests.

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7. $\sum_{n\geq 1}(3x-6)^n/n$, $c_n = 3^n/n$, ratio $(n+1)/(3n)$. (a) Radius. (b) Sum at $x = 11/5$. (c) Interval.

Solution (a): $R = \lim \dfrac{n+1}{3n}$. $\boxed{R = \frac{1}{3}}$

Solution (b): At $x = \frac{11}{5}$, $3x - 6 = \frac{3}{5}$, so the series is $\sum \frac{(3/5)^n}{n} = -\ln\left(1 - \frac{3}{5}\right)$.

$\boxed{\ln \frac{5}{2} \approx 0.916}$

Solution (c) — instructor-judged. With centre 2 and radius $\frac{1}{3}$ the endpoints are $x = \frac{5}{3}$ and $x = \frac{7}{3}$. At $x = \frac{7}{3}$, $3x - 6 = 1$ and the series is harmonic — diverges. At $x = \frac{5}{3}$, $3x - 6 = -1$ and it is alternating harmonic — converges. The interval is $\left[\frac{5}{3}, \frac{7}{3}\right)$. If the factor of 3 is not pulled out, the condition $|3x - 6| < 1$ is misread as $|x - 2| < 1$ and the radius comes out three times too large. Full credit names that trap.

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8. $S(d) = \sum_{n\geq 1}(d-4)^n/(n5^n)$, d in milligrams, ratio $5(n+1)/n$. (a) Radius. (b) $S(6)$. (c) Interval and meaning.

Solution (a): $R = \lim \dfrac{5(n+1)}{n}$. $\boxed{R = 5}$

Solution (b): At $d = 6$, $\dfrac{d-4}{5} = \dfrac{2}{5}$, so $S(6) = \sum \frac{(2/5)^n}{n} = -\ln\left(1 - \frac{2}{5}\right)$. $\boxed{\ln \frac{5}{3} \approx 0.511}$

Solution (c) — instructor-judged. Centre 4, radius 5, endpoints $d = 9$ and $d = -1$. At $d = 9$ the series is harmonic and diverges; at $d = -1$ it is alternating harmonic and converges. The interval of convergence is $[-1, 9)$. Read back into the situation: the model produces a finite index only for doses below 9 milligrams, and since a dose cannot be negative the usable range is $0 < d < 9$ milligrams. Full credit connects the excluded right endpoint to the harmonic series and states the practical range.